

Topic 8 – Fuels and Earth science

Fuels

Students should:	Maths skills
8.1 Recall that hydrocarbons are compounds that contain carbon and hydrogen only	
8.2 Describe crude oil as: a a complex mixture of hydrocarbons b containing molecules in which carbon atoms are in chains or rings (names, formulae and structures of specific ring molecules not required) c an important source of useful substances (fuels and feedstock for the petrochemical industry) d a finite resource	
8.3 Describe and explain the separation of crude oil into simpler, more useful mixtures by the process of fractional distillation	
8.4 Recall the names and uses of the following fractions: a gases, used in domestic heating and cooking b petrol, used as fuel for cars c kerosene, used as fuel for aircraft d diesel oil, used as fuel for some cars and trains e fuel oil, used as fuel for large ships and in some power stations f bitumen, used to surface roads and roofs	
8.5 Explain how hydrocarbons in different fractions differ from each other in: a the number of carbon and hydrogen atoms their molecules contain b boiling points c ease of ignition d viscosity and are mostly members of the alkane homologous series	4a, 4c
8.6 Explain an homologous series as a series of compounds which: a have the same general formula b differ by CH_2 in molecular formulae from neighbouring compounds c show a gradual variation in physical properties, as exemplified by their boiling points d have similar chemical properties	1c, 1d 4a

Students should:	Maths skills
8.7 Describe the complete combustion of hydrocarbon fuels as a reaction in which: a carbon dioxide and water are produced b energy is given out	
8.8 Explain why the incomplete combustion of hydrocarbons can produce carbon and carbon monoxide	
8.9 Explain how carbon monoxide behaves as a toxic gas	
8.10 Describe the problems caused by incomplete combustion producing carbon monoxide and soot in appliances that use carbon compounds as fuels	
8.11 Explain how impurities in some hydrocarbon fuels result in the production of sulfur dioxide	
8.12 Explain some problems associated with acid rain caused when sulfur dioxide dissolves in rain water	
8.13 Explain why, when fuels are burned in engines, oxygen and nitrogen can react together at high temperatures to produce oxides of nitrogen, which are pollutants	
8.14 Evaluate the advantages and disadvantages of using hydrogen, rather than petrol, as a fuel in cars	
8.15 Recall that petrol, kerosene and diesel oil are non-renewable fossil fuels obtained from crude oil and methane is a non-renewable fossil fuel found in natural gas	
8.16 Explain how cracking involves the breaking down of larger, saturated hydrocarbon molecules (alkanes) into smaller, more useful ones, some of which are unsaturated (alkenes)	1c
8.17 Explain why cracking is necessary	2c

Suggested practicals

- Investigate the fractional distillation of synthetic crude oil and the ease of ignition and viscosity of the fractions.
- Investigate the products produced from the complete combustion of a hydrocarbon.
- Investigate the cracking of paraffin oil.

Earth and atmospheric science

Students should:	Maths skills
8.18 Recall that the gases produced by volcanic activity formed the Earth's early atmosphere	
8.19 Describe that the Earth's early atmosphere was thought to contain: a little or no oxygen b a large amount of carbon dioxide c water vapour d small amounts of other gases and interpret evidence relating to this	2c 3a 4a
8.20 Explain how condensation of water vapour formed oceans	
8.21 Explain how the amount of carbon dioxide in the atmosphere was decreased when carbon dioxide dissolved as the oceans formed	
8.22 Explain how the growth of primitive plants used carbon dioxide and released oxygen by photosynthesis and consequently the amount of oxygen in the atmosphere gradually increased	
8.23 Describe the chemical test for oxygen	
8.24 Describe how various gases in the atmosphere, including carbon dioxide, methane and water vapour, absorb heat radiated from the Earth, subsequently releasing energy which keeps the Earth warm: this is known as the greenhouse effect	
8.25 Evaluate the evidence for human activity causing climate change, considering: a the correlation between the change in atmospheric carbon dioxide concentration, the consumption of fossil fuels and temperature change b the uncertainties caused by the location where these measurements are taken and historical accuracy	2c, 2h 4a
8.26 Describe: a the composition of today's atmosphere b the potential effects on the climate of increased levels of carbon dioxide and methane generated by human activity, including burning fossil fuels and livestock farming c that these effects may be mitigated: consider scale, risk and environmental implications	

Use of mathematics

- Extract and interpret information from charts, graphs and tables (2c and 4a).
- Use orders of magnitude to evaluate the significance of data (2h).

Suggested practicals

- Investigate the proportion of oxygen in the atmosphere.
- Investigate the presence of water vapour and carbon dioxide in the atmosphere.
- Investigate the volume of air used up and products formed when candles are burned.
- Carry out the test for oxygen.